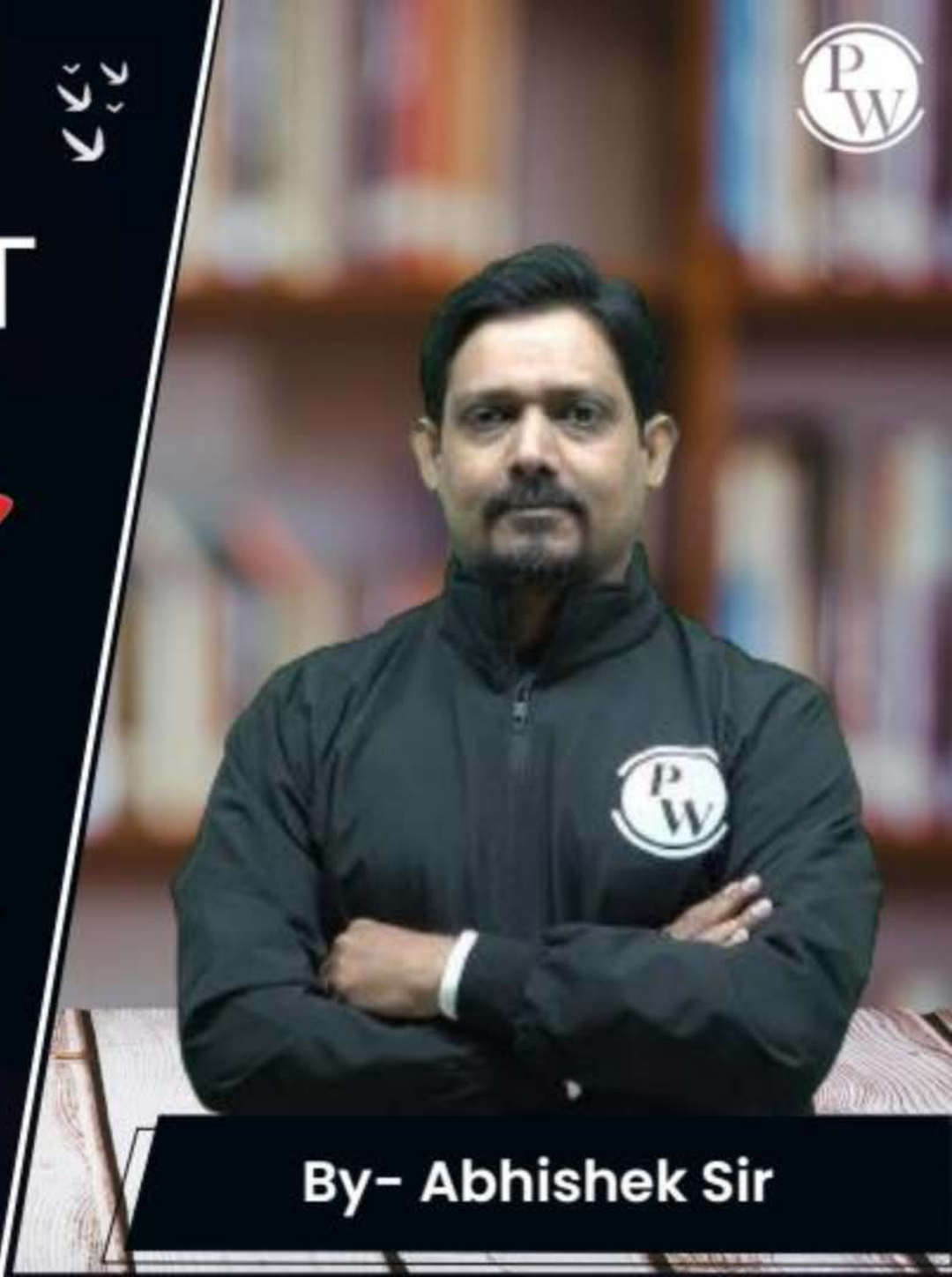
Computer Science & IT



C Programming

String

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Saturday 10 am

declaration of 2-D array

question practice

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

String : Array of character

different ways of declaration of string

Strlen() function

Slide



String

String
Null character
Print string
Two ways to declaration of string
Array of string
How to print array of string
Array of string using character pointer

Data structure

Dynamic Memory Allocation (DMA)

Heap

Structure

Application

Linked list

Tree

Slide



Question

What is printed by the following ANSI C program?

```
#include<stdio.h>
int main(int argc, char *argv[]){
   int x = 1, z[2] = {10, 11};
   int *p = NULL;
   p = &x;
   *p = 10;
   p = &z[1];
   *(&z[0] + 1) += 3;
   printf("%d, %d, %d\n", x, z[0], z[1]);
   return 0;
}
```

(A) 1, 10, 11

(B) 1, 10, 14

(C) 10, 14, 11

(D) 10, 10, 14



Question

What is printed by the following ANSI C program?

```
#include<stdio.h>
int main(int argc, char *argv[])
{
int a[3][3][3] = {{1, 2, 3, 4, 5, 6, 7, 8, 9},
                  {10, 11, 12, 13, 14, 15, 16, 17, 18},
                  19, 20, 21, 22, 23, 24, 25, 26, 27}};
int i = 0, j = 0, k = 0;
for( i = 0; i < 3; i++ ){
     for(k = 0; k < 3; k++ )
          printf("%d ", a[i][j][k]);
     printf("\n");
}
return 0;
}
```

(A) 1 2 3
10 11 12
19 20 21

(B) 1 4 7
10 13 16
19 22 25

(C) 1 2 3
4 5 6
7 8 9

(D) 1 2 3
13 14 15
25 26 27

Slide



Question

Consider the following program

```
#include <stdio.h>
int main(){

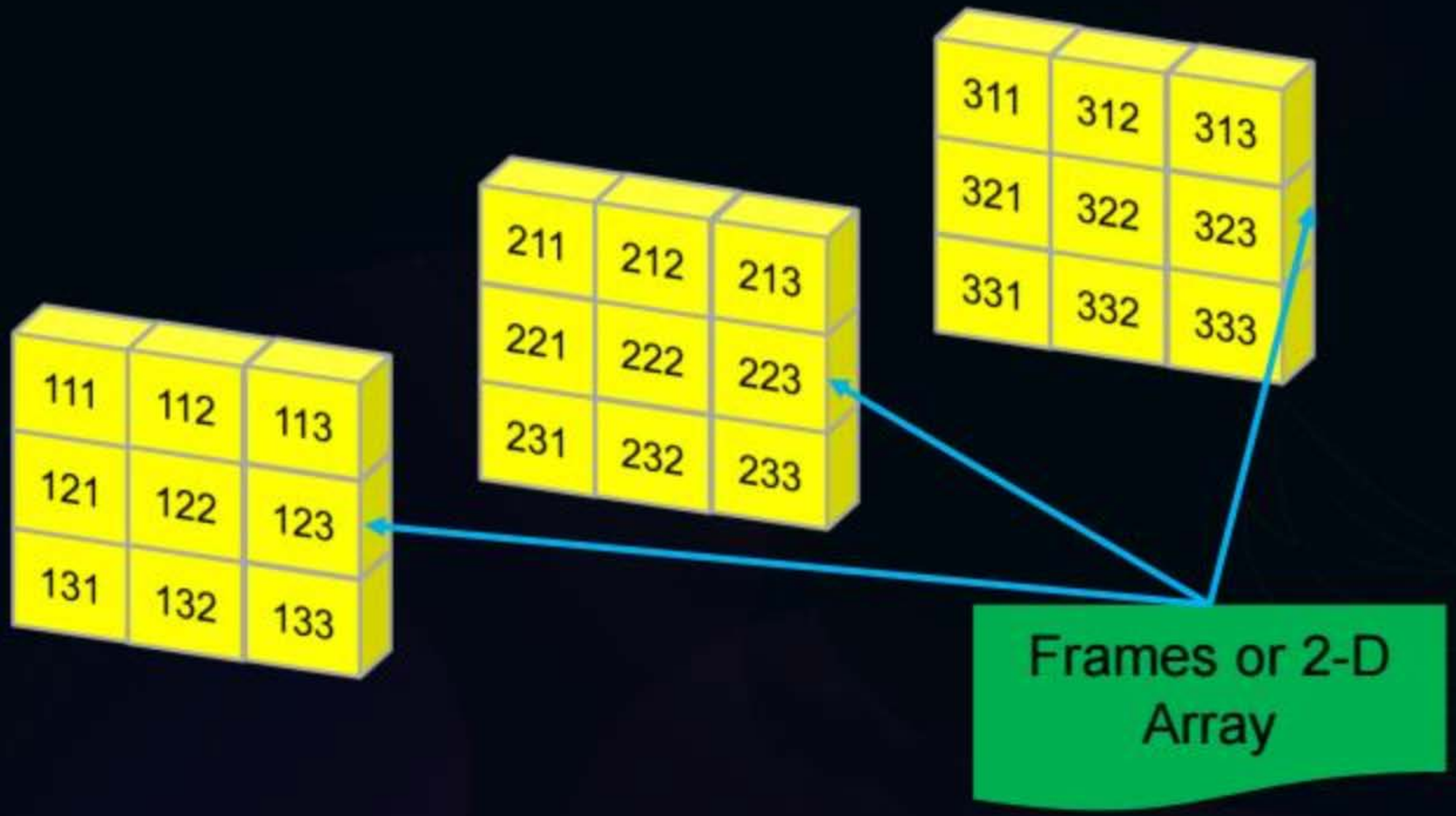
    int a[][2] = {12,34,5,6,78,8,99};
    int *ptr= *(a+1);

    printf ("%d", *ptr++);
    ++*ptr;
    printf ("%d", *ptr);
    return 0;
}
```

The output of the program is

Slide

3-D Array



I will discuss

Satur

Slide



String

String is array of character

printf ("%d", a);

1. printf
2. (
3. "%d"
4. ,
5. a
6.)
7. ;

String token

Any thing between

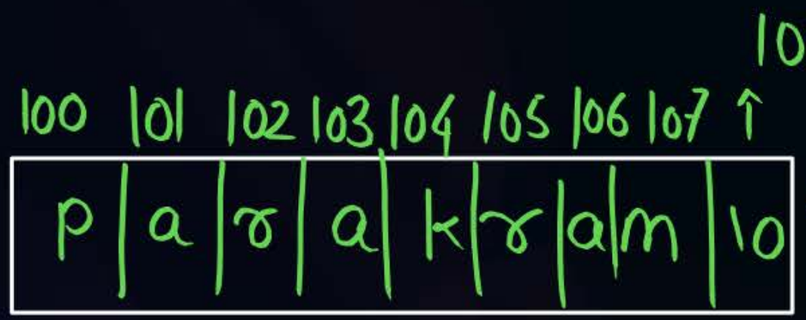
" "

Slide

String

String is array of character

```
char ch[] = "Parakram";
```



To identify End of string NULL character is Inserted at end of string.

Compiler Design

Slide

String

```
Char ch[] = "parakram";
```




```
printf("%c", ch[1]);
```

↑
this will print

```
printf("%c", ch[8]);
```

It will print Nothing

```
printf("%d", ch[8]);
```

this will print 0

Slide

String

```
Char ch[] = "parakram";
```




```
printf("%s", ch);
```

format specifier

Address of first character

this will print : parakram

Slide

String

```
Char ch[] = "parakram";
printf("%s", ch+1);
```

101



It will print : arakram

```
printf("%s", ch+8);
```

← this will print Nothing

Slide



String

first way of declaration & Initialization is using character array

```
char ch[] = "parakram";
```

ch[0]

k / p	a	r	a	k	r	a	m	\0

```
ch[0] = 'k';
```

```
printf("%s", ch);
```

↑ this will print karakram



String

first way of declaration & Initialization is using character array

```
char ch[] = "parakram";
```

ch[0]

p	a	r	a	k	r	a	m	\0

```
ch++;
```

↑ Error (L-value Needed)

Slide



String

first way of declaration & Initialization is using character array

```
char ch[] = "parakram";
```

ch[0]

K	p	a	r	a	k	r	a	m	\0
					10				

```
ch[4] = '\0';
printf("%s", ch);
```

++ ch[i]

Slide

String

first way of declaration & Initialization is using character array

```
char ch[] = "parakram";
++ch[1];
printf("%s", ch);
```

post



pbra

Slide



String

Second way of declaration of string is using character pointer.

Char *ch = "parakram"; ← string will be stored in ROM Read only memory

string if stored using character pointer string is immutable

String

Second way of declaration of string is using character pointer.

```
Char *ch = "prakram";
```

ch | 400

ch++; ← allowed, ch--;

ch[1] = 'K'; ← No Error

printf ("%s", ch); No output



Slide

String

Second way of declaration of string is using character pointer.

```
Char *ch = "prakram";
```

ch | 400

ch++; ← allowed, ch--;

ch[1] = 'K'; ← No Error

printf ("%s", ch); No output



Slide

String

```
char *ch1 = "parakram";
char *ch2 = "parakram";

if (ch1 == ch2)
    printf ("Maza aa gaya");
else
    printf ("C Jindabad");
```



ch1 | 400

ch2 | 400

Maza aa gaya

Slide



String

```
char ch1[]= "parakram";

char ch2[] ="parakram";

if (ch1 == ch2)
        printf ("Maza aa gaya");
else
        printf ("C Jindabad");
```

400





output

C - Jindabad

Slide



String

```
# include< string.h>
# include< stdio.h>

int main() {

    char ch[ ] = "paraakram";

    printf ("%d", (int) strlen(ch));

    return 0;
    }
```

* const

char pointer

return type

unsinged value

↑ type cast

8

Slide



String

```
# include< string.h>
# include< stdio.h>

int main() {

    char ch[ ] = "paraakram";

    printf ("%d", (int) strlen(ch+2));

    return 0;
    }
```

* const char pointer

this will print 6

Slide



String

$ch[1] = *(ch+1) = *(1+ch) = 1[ch];$

$ch[i] = i[ch];$

Slide

Question =

```
#include<stdio.h>
int main(int argc, char *argv[]){
char a = 'P';
char b = 'x';
char c = (a & b) + '*';
char d = (a | b) - '-';
char e = (a ^ b) + '+';
printf("%c %c %c\n", c, d, e);
return 0;
}
```

ASCII encoding for relevant characters is given below

*	+	−
42	43	45

(A) z K S
(B) 122 75 83
(C) * - +
(D) P x +

96
16
112

a = 80
b = 120

01010000

01111000

97 A 65

A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
Q
R
S
T
U
V
W
X
Y
Z

Slide

Question =

```
#include<stdio.h>
int main(int argc, char *argv[]){
char a = 'P';
char b = 'x';
char c = (a & b) + '*';
char d = (a | b) - '-';
char e = (a ^ b) + '+';
printf("%c %c %c\n", c, d, e);
return 0;
}
```

ASCII encoding for relevant characters is given below

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(A) z K S
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Slide

Question =

```
#include<stdio.h>
int main(int argc, char *argv[]){
char a = 'P';
char b = 'x';
char c = (a & b) + '*';
char d = (a | b) - '-';
char e = (a ^ b) + '+';
printf("%c %c %c\n", c, d, e);
return 0;
}
```

ASCII encoding for relevant characters is given below

*	+	−
42	43	45

(A) z K S
(B) 122 75 83
(C) * - +
(D) P x +

Slide

Question

2010

#Q Consider the following C program segment:

```
char p [20];
char *s = "string";
int length = strlen(s);
for (i=0 ; i<length; i++)
        p[i] = s[length - i];
printf("%s",p);
```

String
6

The output of the program is

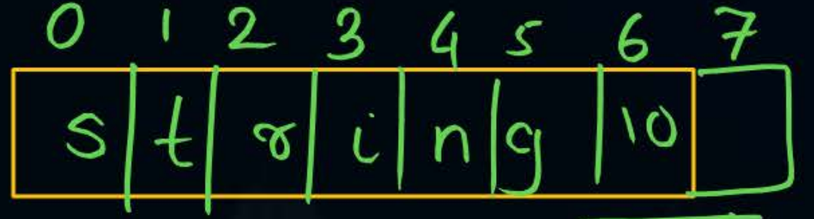
(a) gnirts

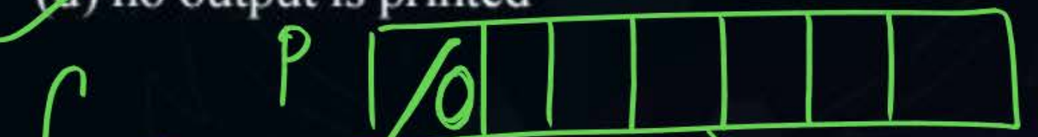
(b) string

(c) gnirt

(d) no output is printed

NULL is not counted





for (i=0; i<6; i++)

P[i] = S[length-i]

P[0] = S[6-0] = S[6]

Slide

Question =

Q What does the following fragment of C-program print?

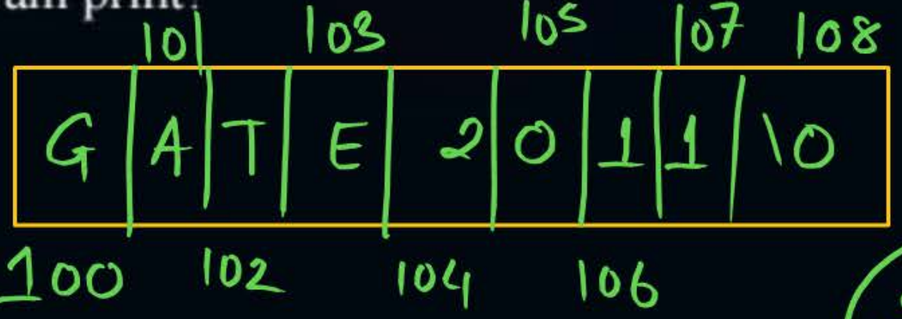
```
char c[]="GATE2011";
char *p =c;
printf("%s", p+p[3]-p[1]);
```

(A) GATE2011

(B) E2011

(C) 2011

(D) 011



2011

p 100

p + p[3] - p[1]

100 + E - A = 100 + 69 - 65

= 104

Slide



Question

#Q Consider the following C Program.

```
#include <stdio.h>
#include< string.h>
int main () {
      char* c = “GATECSIT2017”;
      char* p = c;
      printf(“%d”, (int) strlen (c+2[p]-6[p]-1));
      return 0;
}
```

The output of the program is _____.



Question

Consider the following C program:

```
#include<stdio.h>
void fun1(char *s1, char *s2){
        char *tmp;
        tmp = s1;
        s1 = s2;
        s2 = tmp;
}
void fun2(char **s1, char **s2){
        char *tmp;
        tmp = *s1;
        *s1 = *s2;
        *s2 = tmp;
}
```

```
int main (){
    char *str1 = "Hi",   *str2 = "Bye";
    fun1(str1,  str2);
    printf("%s %s ",  str1, str2);
    fun2(&str1,  &str2);
    printf("%s %s",  str1,  str2);
    return 0;
}
```

(A) Hi Bye Bye Hi

(B) Hi Bye Hi Bye

(C) Bye Hi Hi Bye

(D) Bye Hi Bye Hi

Slide

$a = 10$; Location of variable a is updated with

$a =$

↑ Location ≡ L-value

$b = a$ ← value stored in location

↑ Location



2 mins Summary

t.me/Abhishekshrmapw

Topic

Topic

Topic

Topic

Topic

String

using character array

using character pointer

Strlen() - length. \0 Null character Not Counted

Slide

THANK - YOU

